

AI Usage Note - Test Case Generator Project

Project: Test Case Generator (Infinite Computer Solutions POC)

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1. What AI Helped With

Google Gemini (LLM) was used to generate structured Gherkin test cases from user stories, including positive, negative, and edge scenarios. It also helped in extracting actors, actions, inputs, preconditions, and acceptance criteria from requirements.

AI was also used for story quality analysis (scoring 0–100), test coverage analysis, and identifying missing scenarios or security gaps. Cursor IDE assisted in building Flask APIs, React UI, debugging errors, and generating pytest test cases.

2. What AI Got Wrong (and Fixes)

AI sometimes selected incorrect or unavailable Gemini models, which caused 404 errors. This was fixed by adding fallback models.

Quota limits (429 errors) caused failures during generation, resolved using retry logic and alternative model switching.

LLM outputs were sometimes incomplete (missing negative or edge cases). This was fixed by enforcing minimum scenario rules in prompts.

Formatting issues like markdown code blocks and inconsistent indentation were corrected using post-processing and a Gherkin formatter.

Occasionally, AI generated hallucinated or unrelated scenarios, which were reduced using stricter prompts and QA validation.

3. Best Prompts Used

Prompt 1 - Gherkin Generation:

You are an expert QA engineer. Generate ONLY valid Gherkin. Include at least 3 positive, 3 negative, and 3 edge scenarios. Do not use markdown or explanations.

Prompt 2 - Retry Fix:

Your previous output is incomplete. Regenerate full Feature file with required scenario counts.

Prompt 3 - Story Analysis:

Extract actors, actions, inputs, preconditions, and acceptance criteria. Return ONLY JSON format.

Prompt 4 - Coverage Analysis:

Compare user story and test cases. Return coverage score, missing scenarios, and security gaps in JSON.